Practice Final A

Score / 287

MTH 207 (Fall 2026) — J. Baggett

This is practice, not a preview. The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (287 points against 200) so that it covers more ground, so do not use it to judge timing. The real exam is 120 minutes.

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If $f'(2) = 0$, then f has a highest or lowest value at $x = 2$.

(b) A secant line through two points on the graph of f can be horizontal even when f is never constant.

(c) $\dfrac{d}{dx}\sin\left(x^2\right) = \cos(2x)$.

(d) If $\lim\limits_{x\to a} f(x) = 0$ and $|g(x)| \leq 1$ for all x, then $\lim\limits_{x\to a} f(x)g(x) = 0$.

(e) If f has an absolute minimum value at c, then $f'(c) = 0$.

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(18^{pts}) **2.** Find a point on the curve $x = 4 - y^2$ which is closest to the origin.

(12^{pts}) **3.** For the equation $y^3 - xy = -6$ find the equation of the tangent line to the curve at the point $(1, -2)$.

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(4^{pts}) **4.** State the definition of "f is continuous at a".

(3^{pts}) **5.** Give an example of a function f and an x-value a where f is continuous at a but not differentiable at a.

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(10^{pts}) **6.** Consider the region bounded by $y = 2^x$, $y = 3^x$ and $x = 5$. What is the area of the region?

(6^{pts} ea.) **7.** Evaluate the following limits. As a reminder, justify your steps! Simplify your answer as much as possible. (If the limit does not exist, state why.)

(a) $\lim_{x\to\infty} \dfrac{\sqrt{9x^6 - x}}{x^3 + 1}$

(b) $\lim_{x\to 0} \dfrac{\cos x - 1}{\sin x}$

(c) $\lim_{x\to\infty} \left(1 - \dfrac{1}{x}\right)^x$

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(7^{pts} ea.) **8.** Find the derivatives (with respect to x) of the following functions:

(a) $f(x) = \dfrac{\sin 3x}{\tan^{-1} x}$

(b) $f(x) = \displaystyle\int_3^x \sin(t^9)\, dt$

(c) $f(x) = (\cos x)^{\log_3 x}$

(d) $f(x) = \displaystyle\int_x^2 \cos(t^3)\, dt$

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(8^{pts}) **9.** Given a fixed number $b > 0$, derive the derivative rule for $y = b^x$. (You may use the derivative rule for e^x.)

(7^{pts} ea.) **10.** Evaluate the following indefinite integrals. Simplify your answer as much as possible.

(a) $\displaystyle\int u(u-1)^8\,du$

(b) $\displaystyle\int \frac{5-3x}{1+x^2}\,dx$

(c) $\displaystyle\int \frac{-1}{2^{\sin^{-1}x}\sqrt{1-x^2}}\,dx$

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(8^{pts} ea.) **11.** Evaluate the following definite integrals. Simplify your answer as much as possible.

(a) $\displaystyle\int_1^7 |x-3|\,dx$

(b) $\displaystyle\int_{-1}^{\frac{1}{\sqrt{3}}} \frac{dx}{4x^2+4}$

(c) $\displaystyle\int_0^{81\pi^2} \frac{\sin(\sqrt{x})}{\sqrt{x}}\,dx$

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(10^{pts}) **12.** Choose ONE integral on this page and evaluate it. Clearly mark which integral is your choice! (Suggestion: do your scratch work on the back page or ask for extra paper.)

$$\int \frac{e^{5x}}{e^{10x}+1}\,dx \qquad \int \frac{\sec x + \tan x}{\cos x}\,dx \qquad \int_{-1/3}^{\sqrt{6}/6} \frac{1}{\sqrt{2-9x^2}}\,dx \qquad \int \frac{x}{x^4+2x^2+2}\,dx$$

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(10^{pts}) **13.** Use the definition of the derivative to compute $f'(x)$ if $f(x) = x^2 - 3x$.

(15^{pts}) **14.** Determine $\int_1^5 (3x - 2)\,dx$ as a limit of Riemann sums. (You may use any sampling points you like, but say which you have chosen.)

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(5^{pts}) **15.** Given a function $f(x)$, explain what $\displaystyle\int_a^b f(x)\,dx$ represents in terms of the graph of f.

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(18^{pts}) **16.** A spherical balloon is blown up using an air supply which operates at 2 cm^3/sec. How fast is the radius increasing when the radius is 5 cm?

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(8^{pts}) **17.** Show that the equation $2x + \cos x = 0$ has exactly one root.

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(2^{pts}) **18.** Consider $f(x) = \sqrt{x-2}$ on $[6, 18]$. According to the Mean Value Theorem there is a number c in $(6, 18)$ so that $f'(c) =$ _____. Compute and simplify the number to fill in the blank.

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(5^{pts}) **19.** What is the linearization of $f(x) = x^3$ at 5?

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(8^{pts}) **20.** Find the local minimum(s) of $f(x) = x^4 + 4x^3 + 4x^2 - 1$. Find the local maximum(s).

(10^{pts}) **21.** Use the information given to sketch a graph of a function f

- f is continuous except at $x = -1$
- $f(-3) = -3, f(0) = 0, f(1) = 1, f(2) = 8/9$
- $f'(x) > 0$ on $(-1, 1)$ and $f'(x) < 0$ on $(-\infty, -1) \cup (1, \infty)$
- $f''(x) > 0$ on $(2, \infty)$ and $f''(x) < 0$ on $(-\infty, -1) \cup (-1, 2)$

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(3^{pts} ea.) **22.** An object moves along a horizontal coordinate line in such a way that its position at time t is given by $s(t) = t^3 - 11t^2 + 24t + 100$. Here s is measured in feet and t in seconds.

(a) What is the velocity of the object after 1 second?

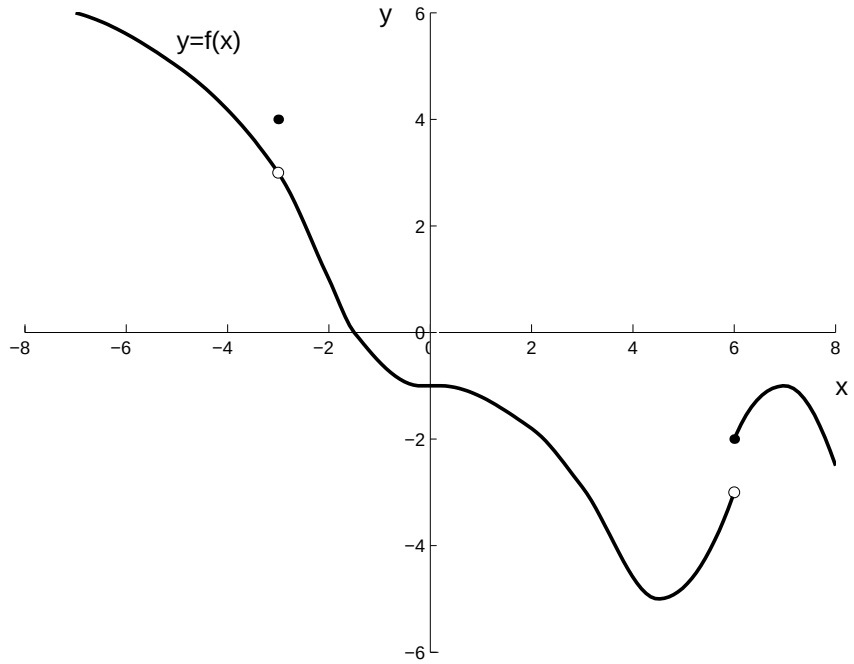
(b) When is the object moving left?

(c) When is the acceleration positive?

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(16[pts]) **23.** From the graph of f, approximate:

Where $f(x) = 0$
Where $f'(x) = 0$
Where f is **not** continuous
Where $f'(x)$ is negative
Where f is **not** differentiable
Inflection point(s) of f
$\lim_{x \to -2} f(x)$
$\lim_{x \to 6^-} f(x)$



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(10^{pts}) **24.** Find the absolute maximum and minimum values of $f(x) = e^x \cos(x)$ on the interval $[0, 2\pi]$. Do NOT use decimal approximations.

THIS IS THE LAST PAGE OF THE EXAM.